Relativity – special and general

In 1905 Albert Einstein showed that time and space are not absolute: moving clocks run slow, and two events that are simultaneous for one observer are not so for another. Ten years later he went further and showed that gravity is the curvature of spacetime. Both theories sounded like philosophy, but today they are engineering: without them GPS would drift by more than ten kilometres a day. This page covers both theories with three interactive figures.

The crisis around 1900

Maxwell's equations predicted that light is an electromagnetic wave with speed $c = 1/\sqrt{\mu_0 \varepsilon_0} \approx 300\,000$ km/s. But speed relative to what? It was assumed that light moves in an ether filling space, and that the Earth rushes through it at about 30 km/s. In 1887 Albert Michelson and Edward Morley built an interferometer that could measure the difference in the speed of light along and across the Earth's motion. They found nothing.

FitzGerald and Lorentz proposed that measuring rods shrink in the direction of motion by exactly enough to hide the effect, and Lorentz and Henri Poincaré found the mathematical transformations that make Maxwell's equations the same for all observers in uniform motion. But they still saw these as effects of an ether.

1905: the two postulates

The 26-year-old patent examiner Albert Einstein in Bern turned the problem on its head. In the paper "On the electrodynamics of moving bodies" (June 1905) he started from just two assumptions:

1. The principle of relativity

The laws of nature are the same for all observers moving uniformly relative to one another. There is no absolute rest and no experiment that can decide who is "really" moving.

2. The speed of light is constant

Light moves with the same speed c in vacuum for all these observers – however the light source moves.

Each postulate is reasonable on its own. Together they are revolutionary, because they can only be reconciled if time and space depend on the observer. The ether became superfluous.

The relativity of simultaneity

Einstein used a train as his example. Lightning strikes both ends of a train passing a platform. An observer on the platform, midway between the strikes, sees the two flashes at the same time and concludes that the strikes were simultaneous. A passenger in the middle of the train is moving towards the light from the front and away from the light from the back; she sees the front flash first. Since light has the same speed for her, she must conclude that the lightning struck the front first. Both are right. Whether two events at different places are simultaneous depends on the motion of the observer.

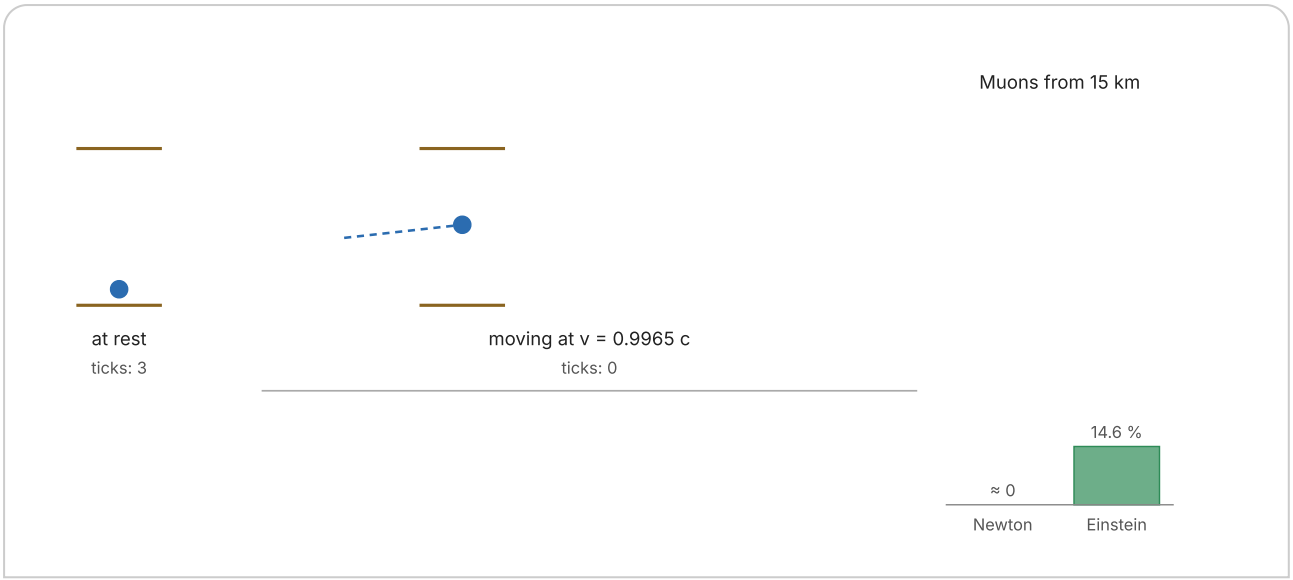
Time dilation and length contraction

Imagine a light clock: a light pulse bouncing up and down between two mirrors. In the clock at rest the light goes straight up and down. Seen by an observer past whom the clock is moving, the light goes on a slant and has further to travel – but at the same speed. Each tick therefore takes longer. Pythagoras gives the time dilation:

$$\Delta t = \gamma \, \Delta t_0, \qquad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}$$

At everyday speeds γ is extraordinarily close to 1. At 90 % of the speed of light it is 2.3, and at 99.5 % it is 10. In the same way a moving object is contracted in the direction of motion by the factor $1/\gamma$.

The effect has been measured directly in nature. Cosmic rays produce muons about 15 km up in the atmosphere. They have a lifetime of only 2.2 µs, and even at almost the speed of light they would on average travel only 660 m before decaying. Yet many of them reach the ground – because their internal clock runs slow. Bruno Rossi and David Hall measured this in 1941, and David Frisch and James Smith repeated it beautifully in 1963 on Mount Washington.



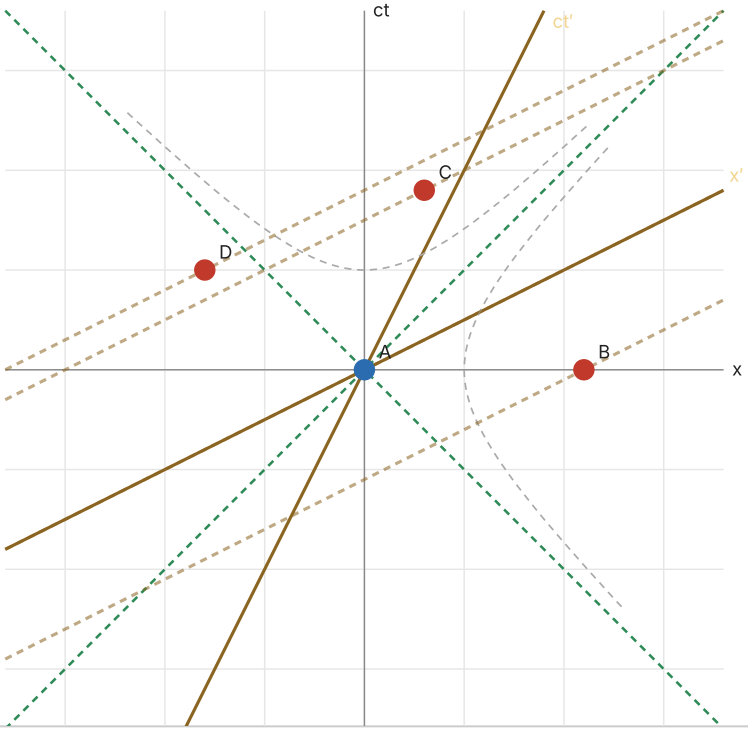
Spacetime

In 1908 Einstein's former mathematics teacher Hermann Minkowski gave a famous lecture in Cologne in which he said that space by itself and time by itself were doomed to fade away into mere shadows, and only a kind of union of the two would preserve an independent reality. He showed that special relativity becomes simple if one thinks of space and time as a single four-dimensional *spacetime*. Observers in different states of motion slice spacetime differently into "space" and "time", but they agree on the spacetime interval:

$$(\Delta s)^2 = (c\,\Delta t)^2 - (\Delta x)^2 - (\Delta y)^2 - (\Delta z)^2$$

The Lorentz transformation is a kind of rotation in spacetime – but with a hyperbolic geometry, in which the axes close up towards the light cone instead of rotating rigidly.

$$x' = \gamma\,(x - vt), \qquad ct' = \gamma\left(ct - \frac{v}{c}x\right)$$



Time in S and S′

	ct	ct′
A	0.00	0.00
B	0.00	−1.27
C	1.80	1.73
D	1.00	2.08

Order on the train (S′):

B → A → C → D

The twin paradox

If time runs slower for whoever is moving, and motion is relative – why is it not symmetric? If one twin travels into space and comes home, she is the younger. The difference is that she changes inertial frame when she turns round; the twin at home does not. In spacetime this corresponds to the travelling twin taking a bent path between the same two events, and in Minkowski's geometry the straight path has the *longest* proper time. In 1971 Joseph Hafele and Richard Keating flew atomic clocks round the world on scheduled flights, eastwards and westwards. The clocks showed exactly the differences that special and general relativity together predicted.

E = mc²

In September 1905 Einstein added a short three-page paper: if a body emits the energy E as radiation, its mass decreases by E/c^2. Mass and energy are two aspects of the same thing:

$$E = mc^2, \qquad E^2 = (pc)^2 + (mc^2)^2$$

The second formula applies to moving particles and to massless particles such as the photon. The formula explains why nuclear reactions release a million times more energy than chemical ones: in fission about a thousandth of the mass is converted into energy (see Nuclear physics). Every second the Sun converts about four million tonnes of mass into radiation.

The equivalence principle

Special relativity did not fit Newton's law of gravitation, in which the force acts instantaneously over any distance. In 1907 Einstein had what he later called the happiest thought of his life: a person falling freely does not feel his own weight. In a freely falling room everything floats as if there were no gravity. Conversely, in a closed box one cannot tell the difference between standing still in a gravitational field and being accelerated by a rocket. This is the *equivalence principle*.

It has two immediate consequences. Light sent horizontally across the rocket will strike slightly lower on the opposite wall – so light must also bend in a gravitational field. And clocks lower down in a gravitational field run slower than clocks higher up. In 1959 Robert Pound and Glen Rebka measured this with gamma rays sent up a 22.5 m tower at Harvard; the frequency difference was $2.5 \cdot 10^{-15}$, exactly as predicted.

Gravity is geometry

If everything falls alike, it is tempting to think of the fall as a property of space and time themselves, not of the bodies. With the help of his fellow student, the mathematician Marcel Grossmann, Einstein learned Riemann's geometry of curved spaces. After eight years of struggle and many mistakes he presented his field equations in November 1915:

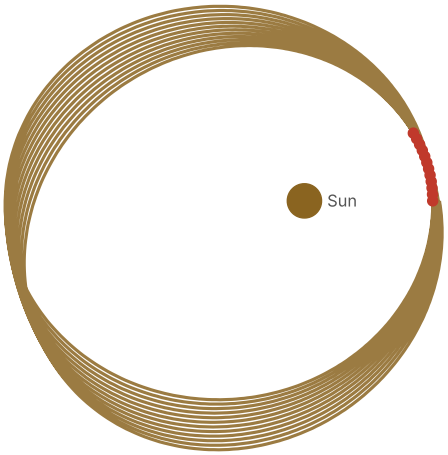
$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

The left-hand side describes the curvature of spacetime, the right-hand side the energy and momentum of matter. John Wheeler summed it up: matter tells spacetime how to curve, and spacetime tells matter how to move. Freely falling bodies follow the "straightest possible" paths in curved spacetime – geodesics. The planets do not orbit the Sun because a force pulls on them, but because they move as straight as spacetime allows. See also Mach's critique of absolute space, which inspired Einstein.

The classical tests

Mercury's perihelion. Mercury's elliptical orbit slowly rotates. Most of this is due to the other planets, but in 1859 Urbain Le Verrier found an excess of about 43 arcseconds per century that no one could explain. Einstein calculated it in November 1915 with his new equations and got exactly 43″. He later wrote that for days he was beside himself with excitement.

$$\Delta\varphi = \frac{6\pi G M_\odot}{c^2 a(1-e^2)} \text{ per orbit} \quad \Rightarrow \quad 43'' \text{ per century for Mercury}$$



The bending of light. Einstein predicted that starlight passing close to the Sun is deflected by 1.75″ – twice as much as a Newtonian calculation gives. During the solar eclipse of 29 May 1919, expeditions led by Arthur Eddington and Frank Dyson measured the positions of the stars on the island of Príncipe and at Sobral in Brazil. The result, published in November 1919, confirmed Einstein and made him world-famous overnight. The measurements were uncertain, but later radio measurements have confirmed the value to far better than a part in a thousand. Today we see gravitational lenses in which entire clusters of galaxies bend the light from more distant galaxies into arcs and rings.

Redshift and delay. The gravitational redshift has been measured in the Pound–Rebka experiment and with atomic clocks that today can measure the difference for a height difference of just one centimetre. In 1964 Irwin Shapiro proposed measuring the delay of radar signals passing close to the Sun – that too agrees.

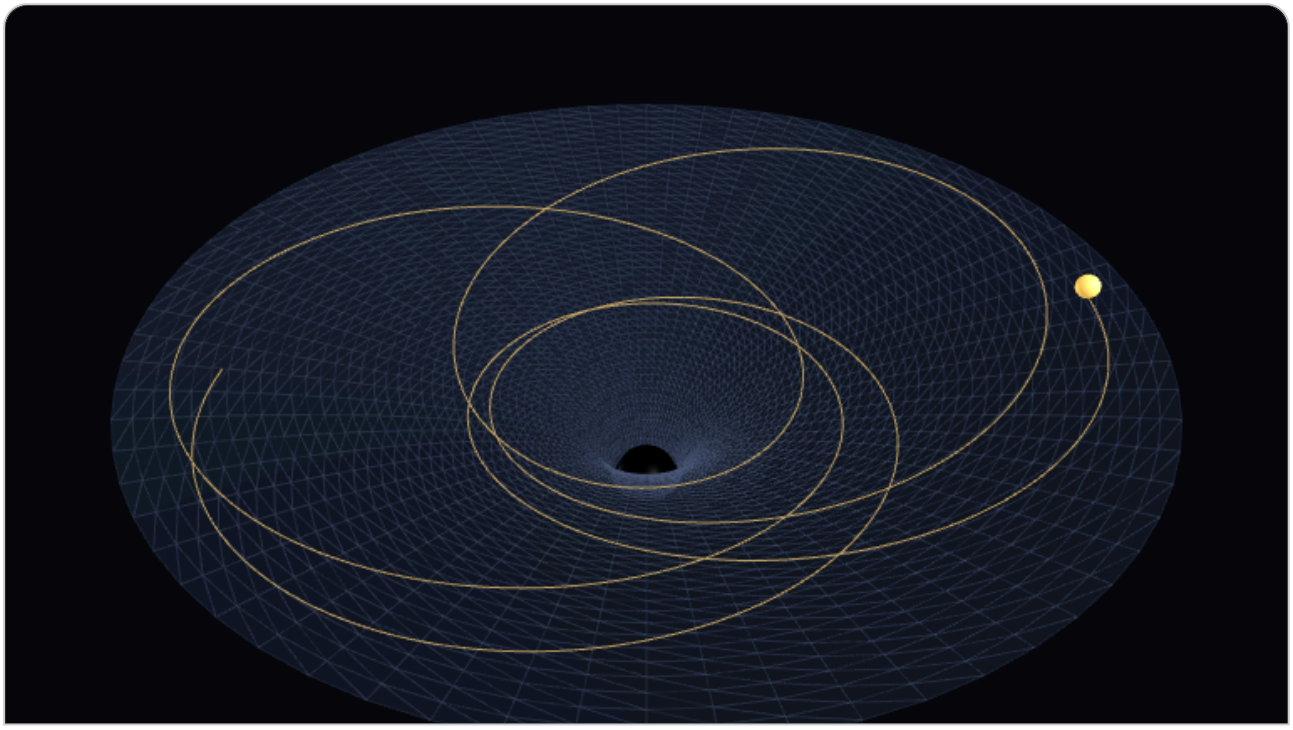
Black holes

As early as December 1915, while serving as a soldier on the Eastern Front, Karl Schwarzschild found the exact solution of the field equations around a spherical mass. It has a strange radius at which time appears to stand still:

$$r_s = \frac{2GM}{c^2} \quad (\approx 3 \text{ km for the Sun}, \;\; \approx 9 \text{ mm for the Earth})$$

It took half a century to understand that this is the horizon of a black hole, and that stars really can collapse into black holes (see Chandrasekhar and Hawking). In 2019 the Event Horizon Telescope showed the first image of the shadow of the black hole in the galaxy M87, and in 2022 of Sagittarius A* at the centre of the Milky Way.

Curved spacetime around a black hole (3D)



Gravitational waves

In 1916 Einstein predicted that accelerated masses emit waves in spacetime, but thought they were far too weak ever to be measured. In 1974 Russell Hulse and Joseph Taylor found a binary pulsar whose orbit shrinks by exactly as much as gravitational waves carry away energy (Nobel Prize 1993). On 14 September 2015 the two LIGO detectors in the USA measured a gravitational wave directly for the first time: two black holes of about 36 and 29 solar masses merging 1.3 billion light years away. Three solar masses were converted into gravitational waves in a fraction of a second, and the four-kilometre arms of the detector were stretched and squeezed by a thousandth of the diameter of a proton. Rainer Weiss, Barry Barish and Kip Thorne received the Nobel Prize in 2017.

GPS: relativity in your pocket

The GPS satellites orbit at a height of 20,200 km at about 3.9 km/s. Their atomic clocks are affected by both theories of relativity:

Effect	Change per day
Special relativity: the satellite's speed makes the clock run slow	approx. −7 µs
General relativity: the weaker gravitational field makes the clock run fast	approx. +45 µs
Total	**approx. +38 µs**

Light travels 11 km in 38 microseconds. Without correction the GPS position would drift by about 11 km a day. The satellite clocks are therefore set to run very slightly slow before they are launched. See also The longitude problem – from Greenwich to GPS.

What is missing?

General relativity has passed every single test for over a hundred years. But it is a classical theory, and it cannot be reconciled with quantum mechanics in any simple way. At the centre of a black hole and at the Big Bang it predicts infinities at which it breaks down. Finding a quantum theory of gravity is one of the greatest open problems in physics – see String theory – and its critics and Nothingness.

Further reading

- Albert Einstein: *Relativity: The Special and the General Theory* (1917, many editions)
- Edwin F. Taylor and John Archibald Wheeler: *Spacetime Physics* (2nd ed. 1992)
- Kip S. Thorne: *Black Holes and Time Warps* (1994)
- Abraham Pais: *"Subtle is the Lord..." The Science and the Life of Albert Einstein* (Oxford 1982)
- Pedro G. Ferreira: *The Perfect Theory* (2014)
- The biographies of Einstein, Lorentz, Mach and Hawking

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